

EnterprateAI Ontology Design

Implementation-Ready Core for Engines, Simulation, ASI, Narratives and PDF Reports

This ontology should be treated as the **business intelligence source of truth** for EnterprateAI, not just a normal SaaS database. The project ontology already defines the correct foundation: Identity, Commercial, Financial, Operational, Market & Growth, and Decision & Simulation domains. It also states that EnterprateAI should treat the business as a full system graph across commercial, financial, operational, market, and decision layers.

1. Core Ontology Principle

Every engine must follow this rule:

Read from ontology

↓

Calculate deterministically

↓

Write structured intelligence back to ontology

↓

Narrative and reports only explain stored intelligence

This allows EnterprateAI to support:

- Viability Engine
- Survival Engine
- Stability Engine
- Growth Engine
- Master Aggregator
- Fragility Index
- Scenario Simulation
- Adaptive Scenario Intelligence
- AI Narratives

- PDF Reports

The ontology document also confirms that simulations should never mutate live ontology state; instead, the platform should store the current state as base, create scenarios and scenario changes, run simulations in memory, and persist only SimulationRun and SimulationResult.

2. High-Level Ontology Structure

Business

- └ Identity Domain
- └ Commercial Domain
- └ Financial Domain
- └ Operational Domain
- └ Market & Growth Domain
- └ Assessment & Intelligence Domain
- └ Fragility Domain
- └ Scenario & Simulation Domain
- └ Adaptive Scenario Intelligence Domain
- └ AI Narrative Domain
- └ Report Domain

3. Recommended Storage

Use **PostgreSQL** as the primary system of record.

Reason:

- relational integrity
- JSONB support
- transactional consistency
- strong event/history support
- scalable enough for MVP and early growth

The existing ontology specifically recommends PostgreSQL and keeping scoring as a separate domain service rather than burying it inside ORM models.

4. Domain A - Identity Domain

Purpose

Stores the core business identity and profile.

Table: businesses

Field	Type	Purpose
id	uuid PK	Business identifier
owner_user_id	uuid FK	Primary owner
name	varchar	Trading name
legal_name	varchar nullable	Registered name
registration_number	varchar nullable	Companies House number
country_code	varchar(2)	GB, NG, US
sector_code	varchar	consulting, retail, SaaS
business_stage	enum	idea, pre_revenue, early_revenue, growth, mature
operating_status	enum	active, paused, closed
base_currency	varchar(3)	GBP
timezone	varchar	Europe/London
created_at	timestamp	Created date
updated_at	timestamp	Updated date

Table: business_profiles

Field	Type	Purpose
id	uuid PK	Profile identifier
business_id	uuid FK unique	Linked business
business_type	enum	service, product, SaaS, marketplace, hybrid

business_model_type	enum	one_off, subscription, recurring, hybrid
delivery_model	enum	manual, hybrid, automated
market_scope	enum	local, regional, national, global
founder_count	int	Founder number
employee_count	int	Team size
digital_component_pct	numeric	Used by Growth Engine
description	text	Business description
notes	text	Internal notes

Used By

Engine/Layer	Use
Growth Engine	delivery model, digital component, market scope
Master Aggregator	business stage
ASI	adaptive recommendations by stage
PDF Reports	business overview
AI Narratives	contextual explanation

5. Domain B - Commercial Domain

Purpose

Stores how the business makes money.

The Viability Engine reads from offerings, pricing models, revenue streams, revenue records, expense records, customers, contracts, opportunities, and demand evidence.

Table: offerings

Field	Type	Purpose
id	uuid PK	Offering identifier
business_id	uuid FK	Owner business
name	varchar	Product/service name

offering_type	enum	product, service, subscription, package
category	varchar	Consulting, training, cleaning, SaaS
description	text	Offering details
unit_name	varchar	hour, project, seat, item
status	enum	draft, active, paused, retired
created_at	timestamp	Created date
updated_at	timestamp	Updated date

Table: pricing_models

Field	Type	Purpose
id	uuid PK	Pricing model
offering_id	uuid FK	Linked offering
pricing_type	enum	fixed, hourly, subscription, usage, tiered
base_price	numeric	Main price
min_price	numeric nullable	Lowest allowed price
max_price	numeric nullable	Highest allowed price
currency	varchar(3)	GBP
price_elasticity_indicator	enum	low, medium, high, unknown
active_from	date	Start date
active_to	date nullable	End date

Table: revenue_streams

Field	Type	Purpose
id	uuid PK	Revenue stream
business_id	uuid FK	Business
offering_id	uuid FK nullable	Linked offering
name	varchar	Revenue stream name
revenue_type	enum	one_off, recurring, contract, usage

recurring_flag	boolean	Recurring revenue
contract_backed_flag	boolean	Contract-backed
active_flag	boolean	Active/inactive

Table: customers

Field	Type	Purpose
id	uuid PK	Customer
business_id	uuid FK	Business
customer_type	enum	individual, startup, SME, corporate
name	varchar	Customer name
segment_id	uuid nullable	Customer segment
acquisition_channel_id	uuid nullable	Channel
country_code	varchar(2)	Country
status	enum	prospect, active, inactive, lost
created_at	timestamp	Created date
updated_at	timestamp	Updated date

Table: contracts

Field	Type	Purpose
id	uuid PK	Contract
business_id	uuid FK	Business
customer_id	uuid FK	Customer
contract_type	enum	one_off, retainer, subscription, framework
start_date	date	Start
end_date	date nullable	End
contract_value	numeric	Total value
recurring_value	numeric nullable	Monthly recurring value
payment_terms_days	int nullable	Payment delay
signed_date	date nullable	Signed date

status	enum	draft, signed, active, expired, cancelled
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Table: opportunities

Field	Type	Purpose
id	uuid PK	Opportunity
business_id	uuid FK	Business
customer_id	uuid nullable	Customer/prospect
offering_id	uuid nullable	Offering
stage	enum	lead, qualified, proposal, negotiation, won, lost
expected_value	numeric	Pipeline value
close_probability	numeric	Probability
expected_close_date	date	Close date
status	enum	open, won, lost

Table: demand_evidence

The Service Idea Viability Engine says idea/service validation should write back structured intelligence including service/offering, revenue assumptions, cost structure, demand evidence, capacity profile, assessment record, scores, outcome, risk flags, and assessment date.

Field	Type	Purpose
id	uuid PK	Evidence ID
business_id	uuid FK	Business
offering_id	uuid FK nullable	Offering
evidence_type	enum	assumption_only, market_research, enquiries, LOIs, paid_pilot, paying_customers, repeat_paying_customers

number_of_interested_leads	int default 0	Lead count
number_of_paying_customers	int default 0	Paying customer count
reference_entity_type	varchar nullable	contract, opportunity, revenue_record
reference_entity_id	uuid nullable	Entity ID
evidence_strength_score	numeric	Demand score
verified_flag	boolean	Verified or estimated
confidence_level	enum	low, medium, high
evidence_date	date	Evidence date

6. Domain C - Financial Domain

Purpose

Stores revenue, expenses, cash, burn, runway and cashflow history.

The Survival Engine requires cash balance, monthly revenue, monthly expenses, variable costs, fixed costs, net cashflow, and historical revenue/expense/cashflow arrays.

Table: financial_periods

Field	Type
id	uuid PK
business_id	uuid FK
period_type	enum: monthly, quarterly, yearly
start_date	date
end_date	date
label	varchar
closed_flag	boolean

Table: revenue_records

Field	Type
id	uuid PK
business_id	uuid FK
period_id	uuid FK
revenue_stream_id	uuid FK nullable
offering_id	uuid FK nullable
customer_id	uuid FK nullable
contract_id	uuid FK nullable
amount	numeric
quantity	numeric nullable
unit_price	numeric nullable
recurring_flag	boolean
source_type	enum: manual, invoice, accounting_sync, bank_sync, imported
confidence_level	enum
recorded_at	timestamp

Table: expense_records

Field	Type
id	uuid PK
business_id	uuid FK
period_id	uuid FK
transaction_id	uuid nullable
expense_type	enum
cost_classification	enum: fixed, variable, mixed
amount	numeric
fixed_flag	boolean
variable_flag	boolean
supplier_id	uuid nullable
source_type	enum
confidence_level	enum
recorded_at	timestamp

Table: cash_accounts

Field	Type
id	uuid PK
business_id	uuid FK
provider_type	enum: manual, bank, wallet, accounting
provider_name	varchar nullable
account_name	varchar
currency	varchar(3)
current_balance	numeric
available_balance	numeric nullable
connected_flag	boolean
last_synced_at	timestamp nullable

Table: cashflow_snapshots

Field	Type
id	uuid PK
business_id	uuid FK
period_id	uuid FK
opening_cash	numeric
cash_inflow	numeric
cash_outflow	numeric
net_cashflow	numeric
closing_cash	numeric
snapshot_date	timestamp
confidence_level	enum

Used By

Engine/Layer	Data Needed
Viability	revenue, variable costs, fixed costs
Survival	cash balance, burn, runway, volatility

Stability	revenue history
Fragility	cash fragility, revenue fragility
Simulation	price/cost/revenue changes
ASI	cash pressure triggers
Reports	cashflow summary

7. Domain D - Operational Domain

Purpose

Stores staff, processes, dependencies, capacity and backlog.

The Stability Engine uses revenue predictability, customer concentration, operational dependency and capacity utilisation.

Table: roles

Field	Type
id	uuid PK
business_id	uuid FK
name	varchar
function_area	varchar
criticality_level	enum: low, medium, high, critical
replaceability_level	enum: easy, moderate, hard, very_hard

Table: staff

Field	Type
id	uuid PK
business_id	uuid FK
name	varchar
staff_type	enum: founder, employee, contractor, advisor
role_id	uuid nullable

monthly_cost	numeric nullable
active_flag	boolean
key_person_flag	boolean
start_date	date nullable
end_date	date nullable

Table: processes

Field	Type
id	uuid PK
business_id	uuid FK
name	varchar
process_type	enum
documented_flag	boolean
automation_level	enum: low, medium, high
owner_role_id	uuid nullable
status	enum: active, weak, missing, deprecated

Table: dependencies

The ontology already includes Dependency as the entity for single-point-of-failure and operational reliance.

Field	Type
id	uuid PK
business_id	uuid FK
dependency_type	enum: key_person, customer, supplier, system, process, channel
source_entity_type	varchar
source_entity_id	uuid
target_entity_type	varchar
target_entity_id	uuid
criticality_level	enum
description	text
active_flag	boolean

Table: capacity_profiles

Field	Type
id	uuid PK
business_id	uuid FK
offering_id	uuid nullable
capacity_unit_name	varchar
max_capacity_units	numeric
current_capacity_units	numeric nullable
current_utilisation_pct	numeric nullable
review_period_type	enum

Table: capacity_snapshots

Field	Type
id	uuid PK
business_id	uuid FK
period_id	uuid FK
capacity_profile_id	uuid FK
available_units	numeric
used_units	numeric
utilisation_pct	numeric
backlog_units	numeric nullable
snapshot_date	timestamp

Table: backlogs

Field	Type
id	uuid PK
business_id	uuid FK
period_id	uuid FK
backlog_type	enum
volume	numeric

estimated_value	numeric nullable
ageing_days	int nullable

8. Domain E - Market & Growth Domain

Purpose

Stores the data needed by Growth Engine and market fragility.

The Growth Engine evaluates scalability potential, market expansion readiness, pricing power and competitive strength.

Table: market_profiles

Field	Type
id	uuid PK
business_id	uuid FK unique
primary_market_scope	enum: local, regional, national, global
geographic_scalability	boolean
regulatory_barriers	enum: low, medium, high
market_maturity	enum: emerging, growing, mature, declining
notes	text

Table: channels

Field	Type
id	uuid PK
business_id	uuid FK
channel_type	enum
name	varchar
active_flag	boolean
scalability_level	enum: low, medium, high

Table: competitors

Field	Type
id	uuid PK
business_id	uuid FK
name	varchar
competitor_type	enum
market_position_estimate	enum
pricing_position	enum
differentiation_notes	text

Table: positioning_profiles

Field	Type
id	uuid PK
business_id	uuid FK unique
differentiation_level	enum: low, medium, high
brand_strength	enum: low, medium, high
switching_cost_level	enum: low, medium, high
market_position	enum: leader, challenger, niche, unknown
pricing_power_level	enum: low, medium, high

Table: growth_signals

Field	Type
id	uuid PK
business_id	uuid FK
signal_type	enum
signal_value	numeric nullable
signal_unit	varchar nullable
signal_date	date
source_type	enum
confidence_level	enum

9. Domain F - Assessment & Intelligence Domain

Purpose

Stores all engine outputs.

This is where the platform becomes intelligence-led.

The ontology already defines Assessment, AssessmentMetric, AssessmentScore, RiskFlag, Interpretation and Recommendation as core decision entities.

Table: assessments

Field	Type
id	uuid PK
business_id	uuid FK
assessment_type	enum
source_context	enum
ruleset_version	varchar
scoring_mode	enum
triggered_by	enum
assessment_date	timestamp
confidence_score	numeric
confidence_level	enum
status	enum
input_snapshot	jsonb
output_snapshot	jsonb

assessment_type enum

viability
survival
stability
growth
master

fragility
simulation
adaptive

Table: assessment_metrics

Field	Type
id	uuid PK
assessment_id	uuid FK
engine_name	enum
metric_name	varchar
metric_value	numeric
metric_unit	varchar nullable
metric_type	enum
explanation	text nullable

Table: assessment_scores

Field	Type
id	uuid PK
assessment_id	uuid FK
engine_name	enum
score_name	varchar
score_value	numeric
weight	numeric nullable
classification_code	varchar nullable

Table: risk_flags

Field	Type
id	uuid PK
business_id	uuid FK
assessment_id	uuid FK nullable
engine_name	enum

flag_code	varchar
risk_category	enum
severity	enum
trigger_condition	text
explanation	text
recommended_action	text
priority_score	numeric
confidence_level	enum
active_flag	boolean
created_at	timestamp
resolved_at	timestamp nullable

Table: recommendations

Field	Type
id	uuid PK
business_id	uuid FK
assessment_id	uuid FK nullable
source_engine	enum
recommendation_type	enum
recommendation_text	text
priority_level	enum
expected_impact_score	numeric nullable
implementation_difficulty	enum nullable
simulation_hint	text nullable
status	enum: open, accepted, rejected, completed
created_at	timestamp

Table: interpretations

Field	Type
id	uuid PK
assessment_id	uuid FK
engine_name	enum

headline	text
driver_summary	text
business_interpretation	text
risk_outlook	text
data_quality_note	text nullable

10. Domain G - Master Aggregator Domain

Purpose

Stores unified business condition.

Table: master_assessments

Field	Type
id	uuid PK
business_id	uuid FK
assessment_id	uuid FK
viability_score	numeric
survival_score	numeric
stability_score	numeric
growth_score	numeric
business_intelligence_score	numeric
master_classification	enum
structural_classification	enum
confidence_score	numeric
confidence_level	enum
created_at	timestamp

Example structural classifications

HIGH_PERFORMING
STRONG_BUT_NEEDS_OPTIMISATION
AT_RISK
CRITICAL
VIABLE_BUT_CASH_CRITICAL
SCALING_WITH_FRAGILITY
STABLE_BUT_STAGNANT
GROWTH_READY_BUT_UNSTABLE

The ontology example assessment response already includes viability, survival, stability, growth and business intelligence score with classifications for each engine and master classification.

11. Domain H - Fragility Index Domain

Purpose

Stores structural vulnerability.

Table: fragility_profiles

Field	Type
id	uuid PK
business_id	uuid FK
assessment_id	uuid FK nullable
master_assessment_id	uuid FK nullable
fragility_index	numeric
classification	enum
cash_fragility	numeric
revenue_fragility	numeric
customer_fragility	numeric

operational_fragility	numeric
capacity_fragility	numeric
market_fragility	numeric
confidence_score	numeric
confidence_level	enum
created_at	timestamp

classification enum

RESILIENT

STABLE_WITH_EXPOSURE

FRAGILE

HIGHLY_FRAGILE

CRITICAL_FRAGILITY

Table: fragility_factors

Field	Type
id	uuid PK
fragility_profile_id	uuid FK
factor_code	varchar
dimension	enum
severity	enum
score_impact	numeric
explanation	text
mitigation_action	text

Table: risk_propagations

Field	Type
id	uuid PK
business_id	uuid FK
source_risk_flag_id	uuid FK nullable
source_entity_type	varchar
source_entity_id	uuid nullable
downstream_entity_type	varchar

downstream_entity_id	uuid nullable
propagation_path	jsonb
propagation_score	numeric
explanation	text
created_at	timestamp

12. Domain I - Scenario & Simulation Domain

Purpose

Stores user-created and system-created scenarios without changing live ontology.

The Simulation Engine requires scenario types such as price change, cost change, hiring, demand drop, automation, capacity increase and risk scenarios. It must reuse the same formulas as Viability, Survival, Stability and Growth, modify only selected variables, and recompute all layers.

Table: scenarios

Field	Type
id	uuid PK
business_id	uuid FK
name	varchar
scenario_type	enum
scenario_source	enum: user, asi, system_preset
based_on_assessment_id	uuid nullable
created_by	uuid nullable
status	enum: draft, ready, executed, archived
created_at	timestamp

Table: scenario_changes

Field	Type
id	uuid PK
scenario_id	uuid FK
entity_type	varchar
entity_id	uuid nullable
field_name	varchar
old_value	text nullable
new_value	text
change_unit	varchar nullable
change_direction	enum: increase, decrease, replace
change_pct	numeric nullable

Table: simulation_runs

Field	Type
id	uuid PK
business_id	uuid FK
scenario_id	uuid FK
base_assessment_id	uuid nullable
run_date	timestamp
status	enum: running, completed, failed
execution_time_ms	int nullable
base_state_snapshot	jsonb
simulated_state_snapshot	jsonb

Table: simulation_results

Field	Type
id	uuid PK
simulation_run_id	uuid FK
engine_name	enum

base_score	numeric nullable
simulated_score	numeric nullable
score_delta	numeric nullable
base_metrics	jsonb
simulated_metrics	jsonb
new_risk_flags	jsonb
removed_risk_flags	jsonb
recommendation_shift	jsonb
summary	text nullable

The Simulation Engine output should compare base scores, simulated scores, score deltas, risk shifts and recommendations.

13. Domain J - Adaptive Scenario Intelligence Domain

Purpose

Stores proactive intelligence generated by the system.

Table: adaptive_triggers

Field	Type
id	uuid PK
business_id	uuid FK
trigger_code	varchar
trigger_type	enum: risk, opportunity, trend, user
source_engine	enum nullable
trigger_condition	text
detected_value	numeric nullable

threshold_value	numeric nullable
severity	enum
confidence_level	enum
status	enum: open, processed, dismissed
created_at	timestamp

Table: adaptive_insights

Field	Type
id	uuid PK
business_id	uuid FK
trigger_id	uuid FK nullable
insight_type	enum
headline	text
insight_summary	text
recommended_decision_path	text
confidence_score	numeric
confidence_level	enum
created_at	timestamp

Table: adaptive_scenario_sets

Field	Type
id	uuid PK
business_id	uuid FK
adaptive_insight_id	uuid FK
objective	enum: improve_survival, reduce_fragility, increase_growth, improve_margin
ranking_method	varchar
best_scenario_id	uuid nullable
created_at	timestamp

Table: adaptive_scenario_rankings

Field	Type
id	uuid PK
scenario_set_id	uuid FK
scenario_id	uuid FK
simulation_run_id	uuid FK nullable
rank_position	int
overall_impact_score	numeric
survival_impact	numeric nullable
fragility_reduction	numeric nullable
growth_impact	numeric nullable
feasibility_score	numeric nullable
confidence_score	numeric
ranking_reason	text

14. Domain K - AI Narrative Domain

Purpose

Stores AI-generated explanations grounded in deterministic outputs.

Table: ai_narratives

Field	Type
id	uuid PK
business_id	uuid FK
source_type	enum: assessment, simulation, fragility, adaptive, report
source_id	uuid

narrative_type	enum: summary, explanation, recommendation, report_section, chat_response
prompt_version	varchar
model_name	varchar
input_context_snapshot	jsonb
generated_text	text
validation_status	enum: passed, failed, needs_review
confidence_level	enum
created_at	timestamp

Table: ai_conversations

Field	Type
id	uuid PK
business_id	uuid FK
user_id	uuid FK
title	varchar nullable
created_at	timestamp
updated_at	timestamp

Table: ai_messages

Field	Type
id	uuid PK
conversation_id	uuid FK
role	enum: user, assistant, system
message_text	text
referenced_assessment_id	uuid nullable
referenced_simulation_run_id	uuid nullable
created_at	timestamp

15. Domain L - PDF Report Domain

Purpose

Stores reports generated from deterministic and narrative outputs.

Table: reports

Field	Type
id	uuid PK
business_id	uuid FK
report_type	enum
title	varchar
status	enum: draft, generating, completed, failed
source_assessment_id	uuid nullable
source_simulation_run_id	uuid nullable
source_fragility_profile_id	uuid nullable
file_url	text nullable
generated_by	uuid nullable
created_at	timestamp
completed_at	timestamp nullable

report_type enum

business_health_report
viability_report
survival_report
stability_report
growth_report
fragility_report
simulation_report
adaptive_decision_report
investor_summary

Table: report_sections

Field	Type
id	uuid PK
report_id	uuid FK
section_key	varchar
section_title	varchar
content_text	text
source_narrative_id	uuid nullable
section_order	int
created_at	timestamp

16. Engine Read/Write Map

Layer	Reads From	Writes To
Viability Engine	offerings, pricing_models, revenue_records, expense_records, demand_evidence, capacity_profiles	assessments, metrics, scores, risks, recommendations
Survival Engine	cash_accounts, revenue_records, expense_records, cashflow_snapshots	assessments, metrics, scores, risks, recommendations
Stability Engine	revenue_records, customers, contracts, dependencies, staff, processes, capacity_snapshots, backlogs	assessments, metrics, scores, risks, recommendations
Growth Engine	business_profiles, market_profiles, positioning_profiles, competitors, channels, growth_signals, pricing_models	assessments, metrics, scores, risks, recommendations
Master Aggregator	assessment_scores, risk_flags, recommendations	master_assessments, consolidated risks
Fragility Index	master_assessments, risk_flags, dependencies, cashflow, customer concentration, capacity	fragility_profiles, fragility_factors, risk_propagations
Scenario Simulation	all score-relevant entities, scenarios, scenario_changes	simulation_runs, simulation_results

ASI	master_assessments, fragility_profiles, risk_flags, simulation_results	adaptive_triggers, adaptive_insights, scenario_sets, rankings
AI Narrative	assessments, simulations, fragility, adaptive insights	ai_narratives
PDF Reports	assessments, narratives, simulations, fragility	reports, report_sections

17. Minimum MVP Ontology Tables

Start with these first:

businesses
business_profiles
offerings
pricing_models
customers
contracts
opportunities
demand_evidence
financial_periods
revenue_records
expense_records
cash_accounts
cashflow_snapshots
staff
roles
processes
dependencies
capacity_profiles
capacity_snapshots
backlogs
market_profiles
channels
competitors
positioning_profiles
growth_signals

assessments
assessment_metrics
assessment_scores
risk_flags
recommendations
interpretations
master_assessments
fragility_profiles
fragility_factors
scenarios
scenario_changes
simulation_runs
simulation_results
adaptive_triggers
adaptive_insights
adaptive_scenario_sets
adaptive_scenario_rankings
ai_narratives
reports
report_sections

18. Most Important Implementation Rule

Do not allow each engine to invent its own database structure.

All engines should use:

- assessment_metrics
- assessment_scores
- risk_flags
- Recommendations
- interpretations

That gives you one standard intelligence language across the whole platform.

19. Final Architecture Verdict

This ontology is the correct foundation for EnterprateAI because it supports:

- structured business modelling
- deterministic scoring
- historical intelligence
- simulation without mutating live data
- fragility analysis
- adaptive intelligence
- grounded AI narratives
- investor/report-ready PDF outputs

The most important thing is this:

The ontology must become the memory of the business, while the engines become the reasoning system.

That is what makes EnterprateAI more valuable than a normal AI assistant or dashboard.